LAB ASSIGNMENT – 3

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Course: B-TECH (CSE)

Section: NA (L1)

College: GLA University

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1. Create a BMI Calculator Using JavaScript

Aim: Calculate BMI using height and weight and display Underweight, Normal, or Overweight.
Formula: BMI = weight (kg) / height (m)². Enter height in centimetres.

Code:

```
<!DOCTYPE html>
<html>
<body>
<h2>BMI Calculator</h2>
Weight (kg): <input type="number" id="w"><br><br>
Height (cm): <input type="number" id="h"><br><br>
<button onclick="calc()">Calculate BMI</button>
<p id="result"></p>
<script>
function calc() {
const w = Number(document.getElementById("w").value);
const h = Number(document.getElementById("h").value) / 100;
const r = document.getElementById("result");
if (w <= 0 || h <= 0) {
r.textContent = "Enter valid height and weight.";
return;
}
const bmi = w / (h * h);
let category = bmi < 18.5 ? "Underweight" :
bmi < 25 ? "Normal" : "Overweight";
r.textContent = "BMI: " + bmi.toFixed(2) + " - " + category;
}
</script>
</body>
</html>
```

Result: The BMI value and category are displayed after clicking Calculate BMI.

2. Make a Signup Page and Apply Form Validations

Aim: Create a signup form and check that fields are completed, email format is valid, password has at least 8 characters, and confirmation matches.

Code:

```
<!DOCTYPE html>
<html>
<body>
<h2>Sign Up</h2>
<form id="signup">
Name: <input id="name" type="text"><br><br>
Email: <input id="email" type="text"><br><br>
Password: <input id="pass" type="password"><br><br>
Confirm Password: <input id="confirm" type="password"><br><br>
<button type="submit">Create Account</button>
</form>
<p id="msg"></p>
<script>
document.getElementById("signup").addEventListener("submit", function(e) {
e.preventDefault();
const name = document.getElementById("name").value.trim();
const email = document.getElementById("email").value.trim();
const pass = document.getElementById("pass").value;
const confirm = document.getElementById("confirm").value;
const msg = document.getElementById("msg");
if (!name || !email || !pass || !confirm) {
msg.textContent = "Please fill in every field.";
} else if (!/^[^\s@]+@[^\s@]+\.[^\s@]+$/.test(email)) {
msg.textContent = "Enter a valid email address.";
} else if (pass.length < 8) {
msg.textContent = "Password must be at least 8 characters.";
} else if (pass !== confirm) {
msg.textContent = "Passwords do not match.";
} else {
msg.textContent = "Form validated successfully!";
}
});
</script>
</body>
</html>
```

Result: Validation messages appear for invalid entries. A real signup system also needs server-side validation.

3. Create a Page in Which User Can Change Color Theme

Aim: Allow the user to switch the webpage between light, dark, and blue themes.

Code:

```
<!DOCTYPE html>
<html>
<body id="page">
<h2>Change Color Theme</h2>
<button onclick="setTheme('light')">Light</button>
<button onclick="setTheme('dark')">Dark</button>
<button onclick="setTheme('blue')">Blue</button>
<script>
function setTheme(theme) {
const page = document.getElementById("page");
if (theme === "dark") {
page.style.backgroundColor = "#222";
page.style.color = "white";
} else if (theme === "blue") {
page.style.backgroundColor = "#dbeafe";
page.style.color = "#1e3a8a";
} else {
page.style.backgroundColor = "white";
page.style.color = "black";
}
}
</script>
</body>
</html>
```

Result: Clicking a button changes the page's background and text colors.

4. Create a Digital Clock Using JavaScript

Aim: Display the current time in HH:MM:SS format and update it every second.

Code:

```
<!DOCTYPE html>
<html>
<body>
<h2>Digital Clock</h2>
<h1 id="clock">00:00:00</h1>
<script>
function updateClock() {
const now = new Date();
const h = String(now.getHours()).padStart(2, "0");
const m = String(now.getMinutes()).padStart(2, "0");
const s = String(now.getSeconds()).padStart(2, "0");
document.getElementById("clock").textContent = h + ":" + m + ":" + s;
}
updateClock();
setInterval(updateClock, 1000);
</script>
</body>
</html>
```

Result: The current system time appears on the page and refreshes every second.

5. Create a To-Do List App

Aim: Make an app where users can add tasks, mark them completed, and delete them.

Code:

```
<!DOCTYPE html>
<html>
<head>
<style>
.completed { text-decoration: line-through; color: gray; }
li { margin: 8px 0; }
</style>
</head>
<body>
<h2>To-Do List</h2>
<input id="task" placeholder="Enter a task">
<button onclick="addTask()">Add</button>
<ul id="list"></ul>
<script>
function addTask() {
const input = document.getElementById("task");
const value = input.value.trim();
if (!value) {
alert("Please enter a task.");
return;
}
const li = document.createElement("li");
const text = document.createElement("span");
text.textContent = value;
text.onclick = () => text.classList.toggle("completed");
const del = document.createElement("button");
del.textContent = "Delete";
del.onclick = () => li.remove();
li.appendChild(text);
li.appendChild(document.createTextNode(" "));
li.appendChild(del);
document.getElementById("list").appendChild(li);
input.value = "";
}
</script>
</body>
</html>
```

Result: Users can add tasks, click a task to mark it complete, and delete tasks.